

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SOFTWARE ENGINEERING [DS_3152]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) How can computer systems engineering be distinguished from software engineering in terms of their focus, scope, and application? Illustrate your explanation with examples of product development projects where systems engineering would be most suitable. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Analyze the given program and construct its control flow graph representation. (5)
- ```
void verify_login(char username[], char password[]) {
 int valid_user = 0;
```

```
 if (strcmp(username, "admin") == 0 && strcmp(password, "1234") == 0) {
 100: valid_user = 1;
 goto step110;
 }
 else if (strcmp(username, "guest") == 0)
 goto step100; // allow guest login
 else
 goto step120; // invalid credentials
```

```
step110:
 if (valid_user == 1)
 printf("Access Granted. Welcome, %s\n", username);
 goto step130;
```

```
step120:
 printf("Error: Invalid Username or Password.n");
```

```
step130:
 printf("Login process completed.n");
}
```

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Explain the V-model of the Software Development Life Cycle (SDLC) and demonstrate how its structure and activities can be applied effectively in the development of safety-critical systems. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Explain the concepts of functional and non-functional requirements in a software system. Using your understanding, identify 3 functional requirements and one non-functional requirement for a
- (5)

bank ATM system.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Demonstrate your understanding of a Data Flow Diagram (DFD) by explaining its importance and primitive symbols. Apply these concepts to design a schematic diagram showing both synchronous and asynchronous data flows used to validate a user-entered number in a software application. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Analyze the characteristics of high-level and detailed designs, and determine the specific documents that should be produced upon the completion of each design stage. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Apply your understanding of Use Case Modelling to the Health Monitoring System that tracks a patient's vital signs (heart rate, blood pressure, oxygen level) through wearable sensors and sends alerts to doctors in case of abnormalities. (5)
- A)

Construct a schematic Use Case Diagram showing system interactions using appropriate «include» and «extend» relationships.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) An Online Shopping System enables users to log in, browse products, add selected items to their cart, make secure payments, and receive an order confirmation. The key operations of the system include: user authentication, product selection, cart management, payment processing, and order confirmation. (5)
- Using the concepts of UML Activity Diagrams, apply your understanding to illustrate the complete workflow of the online shopping process. Draw a schematic Activity Diagram showing the control flow from login to order confirmation, including decision nodes, synchronization bars, and start/end states.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Consider the following C function named sort. (5)
- A)
- ```
/* sort takes an integer array and sorts it in ascending order */
void sort(int a[], int n){
    int i,j;
    for(i=0;i< n-1;i
    for(j=i+1;j< n;j++)
    if(a[i]>a[j])
    {
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
    }
}
```

Apply your knowledge of white-box testing to design appropriate test suites for the following techniques:

1. Condition Coverage
2. Path Coverage

Course Outcome: "CO5"

- B) Consider the following C function named sort. (5)
- ```
/* sort takes an integer array and sorts it in ascending order */
void sort(int a[], int n){
```

```

int i,j;
for(i=0;i< n-1;i
for(j=i+1;j< n;j++)
if(a[i]>a[j])
{
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
}

```

Explain the concept of cyclomatic complexity as a software metric, and calculate the cyclomatic complexity for the given program.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

| Event Name | Subject |
|------------|---------|
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058-END SEMSTER-ODD-2025 DS\_3152:SOFTWARE ENGINEERING

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 10        |
| 3            | 3              | CO 3        | 10        |
| 4            | 4              | CO 4        | 10        |
| 5            | 5              | CO 5        | 10        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
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058-END SEMSTER-ODD-2025 DS\_3152:SOFTWARE ENGINEERING

| #     | Level       | No of Questions | Marks (%)  |
|-------|-------------|-----------------|------------|
| 1     | Remember    | 0               | 0(0%)      |
| 2     | Understand  | 0               | 0(0%)      |
| 3     | Application | 4               | 20(44.44%) |
| 4     | Analysis    | 5               | 25(55.56%) |
| 5     | Create      | 0               | 0(0%)      |
| 6     | Evaluate    | 0               | 0(0%)      |
| Total |             | 9               | 45 (100%)  |